EEG Seizure Detection: A Comparative Study of LSTM, CNN-LSTM, Transformer, and State-Space Architectures

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> This document is a paper-ready methodology + results synthesis. Every architectural and training claim is cited to source (`file:line`). Numeric results are sourced from on-disk artifacts only; checkpoints that lack persisted test metrics are flagged explicitly as gaps to re-run.

Abstract

We present a unified comparative study of 15+ neural architectures for automated seizure detection on scalp EEG, evaluated on the CHB-MIT pediatric corpus. All models share a common preprocessing, windowing, and subject-independent splitting pipeline (`src/data_loader/`), enabling an apples-to-apples comparison across four architecture families: (i) recurrent models (Vanilla LSTM, BiLSTM, Attention-BiLSTM, Feature-BiLSTM), (ii) convolutional–recurrent hybrids (CNN-LSTM with multi-scale temporal kernels), (iii) self-attention / vector-quantised transformers (VQ-Transformer "M7"), and (iv) state-space and pretrained foundation models (EEGMamba, BENDR, EEGPT). Training uses Focal Loss with positive-class re-weighting, AdamW with warmup-cosine learning-rate scheduling, threshold tuning via Youden's J or F1-max, and ensemble voting for the HF-model family. On the legacy baseline evaluation the CNN-LSTM hybrid achieves F1 = 0.518 / AUC = 0.712 with sensitivity = 0.569, substantially outperforming the pure LSTM (F1 = 0.346, sensitivity = 0.314) and motivating the later additions of multi-scale CNN frontends, hierarchical context, and pretrained encoders. This document unifies the methodology, hyperparameters, stability fixes, and iteration history into a single reference.

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1. Introduction

Seizure detection from scalp EEG is a dense temporal-classification problem characterised by (a) severe class imbalance (seizures are rare), (b) inter-subject heterogeneity (age, montage, seizure type), (c) non-stationary signal statistics, and (d) a strong demand for subject-independent generalisation in clinical deployment. This codebase tackles the problem with a unified pipeline feeding a wide portfolio of architectures. The explicit research question we attempt to answer: *how do different inductive biases (recurrence, multi-scale convolution, self-attention, state-space dynamics, large-scale self-supervised pretraining) compare when held to the same preprocessing, labelling, and evaluation protocol?*

Scope and contributions: 1. **A unified, subject-independent pipeline** (CHB-MIT → 16-channel 10-20 montage, 256 Hz, 1-s windows) implemented in `src/data_loader/` . 2. **15+ architectures** under one repo with shared loss/metrics utilities (`src/models/utils/`). 3. **A documented stability history** — a sequence of fixes for NaN losses, FP16 underflow, gradient amplification in state-space layers, and class-weighting semantics — which constitutes an engineering contribution in its own right. 4. **Ensemble evaluation** (`ensemble_hf.py`) with per-ensemble threshold tuning.

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2. End-to-End Modeling Pipeline

All models consume batches produced by `data_loader.dataset.factory.create_pytorch_dataloaders()` , which composes the stages below.

2.1 Pipeline diagram

```
PhysioNet CHB-MIT EDF
        │   (download.py:50-85)
        ▼
   MNE read_raw             ──► sfreq, n_channels, duration
        │   (io.py:75-81)
        ▼
   preprocess()
   pick-EEG → resample(256Hz, polyphase) → bandpass 1-50Hz
   → notch 60Hz → average reference
        │   (signal.py:12-82)
        ▼
   standardize_channels()   → 16-ch 10-20 montage
        │   (channels.py:39-138)
        ▼
   BIDS rewrite             → sub-XX/ses-001/eeg/*.edf + sidecars
        │   (bids.py:25-…)
        ▼
   build_window_index()     → 1-s windows, 1-s stride
   label_windows()          → positive if ≥50% overlap with seizure event
        │   (labels.py:11-181)
        ▼
   stratify_subjects()      → 70/15/15 subject-independent splits
        │   (stratify.py:9-84)
        ▼
   balance_index() [train only] → oversample to 30 % positive
        │   (labels.py:184-210)
        ▼
   EEGDataset.__getitem__() → per-channel z-score + random augmentations
        │   (loaders.py:57-60, augment.py:9-68)
        ▼
   PyTorch DataLoader       → (B, 16, 256), (B,)
```

2.2 Data acquisition

Source: CHB-MIT Scalp EEG Database (PhysioNet). Subject metadata (age, sex, 23 recordings / 22 subjects, ages 1.5–22) is hard-coded in `data_loader/core/download.py` (`CHBMIT_SUBJECTS` , lines 22–47); downloader `download_chbmit()` at lines 50–85 fetches all 664 EDF files into a local cache. File reading is handled by `read_raw()` at `data_loader/core/io.py:75–81` via MNE-Python (supports `.edf` , `.bdf` , `.vhdr` , `.set` , `.fif`).

2.3 Preprocessing (`core/signal.py`)

`preprocess()` (lines 12–18) chains: 1. **EEG-channel pick** (`_pick_eeg` , 21–32): MNE channel type filter with fallback to "all channels" when `ch_types` are unset. 2. **Resample to 256 Hz** (`_resample` , 35–46) via polyphase filtering (`_resample_polyphase` , 49–62) — the default path; SciPy-style GCD decomposition minimises phase distortion relative to naive linear interpolation. 3. **Bandpass 1–50 Hz** and **notch at 60 Hz** (`_filter` , 65–73). 4. **Average reference** (`_reference` , 76–82).

Per-channel z-score normalisation (`signal.py:85–90`) is deliberately deferred to the dataset `__getitem__` (`loaders.py:57–60`) so statistics are computed per window, avoiding leakage across train/val/test boundaries.

2.4 Channel standardisation (`core/channels.py`)

Target montage is the 16-channel subset of the 10-20 system: Fp1, Fp2, F3, F4, F7, F8, Fz, Cz, T7, T8, P7, P8, C3, C4, O1, O2 (lines 9–10). `standardize_channels()` (39–52) reduces over-count recordings by case-insensitive name match then spatial nearest-neighbour (`_reduce_to_target` , 55–87) and expands under-count recordings via duplication (`_expand_to_target` , 108–126), finishing with a canonical reorder (129–138). Spatial matching uses a hard-coded `POSITIONS_1020` dict (12–36).

2.5 Windowing & labelling (`core/labels.py`)

`build_window_index()` (153–181) produces non-overlapping 1-s windows at 256 Hz (⇒ 256 samples/window). `label_windows()` (11–42) tags a window positive if its temporal overlap with any seizure event is at least `labeling.overlap_threshold` (default 0.5). Seizure events are parsed from BIDS-style `_events.tsv` companions (107–133) using keyword match against { `seizure` , `sz` , `ictal` , `epilep` }. Negative windows within `windowing.exclude_negatives_within_sec = 300 s` of any seizure are discarded to prevent borderline-labelled background from contaminating the negative class (36–37).

2.6 Subject-independent stratified split (`core/stratify.py`)

`stratify_subjects()` (9–84) bins subjects by age (0–5, 5–10, 10–15, 15–20, 20–30, 30+), tracks seizure status per stratum, and greedily allocates subjects into train/val/test at the 70/15/15 ratio while preserving both age- and seizure-prevalence distributions. `assign_split_column()` (126–149) writes the split back onto the window index so no window can leak across splits.

2.7 Class-imbalance handling at dataset level (`core/labels.py:184–210`)

`balance_index()` oversamples positives on the **train split only** until the positive ratio meets `balance.seizure_ratio = 0.3`. Validation and test splits retain the natural distribution so evaluation is unbiased.

2.8 Augmentation (`core/augment.py:9–68`)

Four stochastic augmentations applied at load time: - **Time warp** (cubic-spline, $\sigma = 0.15$) — non-linear time distortion. - **Per-channel magnitude scale** ($\sigma = 0.15$) — amplitude variation. - **Additive Gaussian noise** at 25 dB SNR. - **Time shift** up to ±15 samples (~59 ms at 256 Hz).

Seizure windows are augmented with probability 0.8 and background with 0.3, biasing augmentation toward the minority class.

2.9 Two parallel data subsystems

The repo currently hosts both `src/dataloaders/` (legacy, `dataloaders/common/{windowing,loader,splits}.py`) and `src/data_loader/` (modern, YAML-configured, BIDS-aware). Both honour identical output contracts. The legacy path is preserved for reproducibility of baseline results on disk; new work targets `data_loader/`. A future revision should collapse the two.

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3. Model Architectures

All models are defined under `src/models/`. The master inventory is in Table 1; per-family detail follows.

Table 1 — Architecture inventory

#	Name	Family	File:line	~Params	Input	Key feature
1	`VanillaLSTM`	LSTM	`architectures/vanilla_lstm.py:47–122`	180 K	16×256	ChannelAttn + uni-LSTM
2	`BiLSTM`	BiLSTM	`architectures/bilstm.py:44–113`	300 K	16×256	ChannelAttn + BiLSTM
3	`AttentionBiLSTM`	BiLSTM + MHA	`architectures/attention_bilstm.py:19–130`	330 K	16×256	Learned pos-emb + 4-head self-attn
4	`FeatureBiLSTM`	BiLSTM on features	`architectures/feature_bilstm.py:19–139`	280 K	226×10	Residual proj + temporal MHA
5	`CNNLSTM`	CNN + BiLSTM	`architectures/cnn_lstm.py:74–186`	480 K	16×256	3-kernel CNN (3/15/31) + BiLSTM + MHA
6–11	`M1…M6`	LSTM benchmark suite	`lstm_benchmark_models/architectures/m[1–6]_*.py`	180–480 K	16×256	Channel / criss-cross / graph attn variants
12	`HierarchicalLSTM`	2-level BiLSTM	`improved_lstm_models/architectures/hierarchical_lstm.py:60–136`	320 K	60×16×256	Window encoder → cross-window BiLSTM
13	`M7_VQTransformer`	Patch-Transformer + VQ	`ensemble_transformers/architectures/m7_vq_transformer.py:13–104`	420 K	16×256	Patch embed + 4-layer Transformer + 512-code VQ
14	`EEGMamba`	SSM	`hugging_face_mamba_moe/architectures/eeg_mamba.py:21–80`	280 K	16×256	Bidirectional Mamba blocks
15	`EEGMambaMoE`	SSM + MoE	`hugging_face_mamba_moe/architectures/eeg_mamba.py`	~320 K	16×256	Mamba + mixture-of-experts router
16	`BENDRPretrainedModel`	Pretrained CNN	`hugging_face_mamba_moe/architectures/pretrained/bendr.py:11–60`	~M (frozen)	16×256	Braindecode BENDR backbone
17	`EEGPTPretrainedModel`	Pretrained Transformer	`hugging_face_mamba_moe/architectures/pretrained/eegpt.py:11–57`	~25 M (frozen)	16×256	`chs_info=None` adapter head
18	`TinySeizureNet`	Depthwise-separable CNN	`approach3/architectures/tiny_seizure_net.py`	< 1 M	16×256	Distilled from M1–M6 teachers

3.1 LSTM family — shared blueprint

All LSTM models follow the pattern `ChannelAttention → LayerNorm → Linear(16→128) → LSTM → Dual(Mean+Max)Pool → FC`. Design rationale:

- **ChannelAttention (`lstm_benchmark_models/modules/channel_attention.py`)** — squeeze-and-excitation over channels. EEG channels carry very different amounts of seizure information; a learned gating vector suppresses noisy or artifact channels before temporal processing.
- **Input projection to 128-dim hidden space** — decouples LSTM capacity from raw channel count (important because the pipeline can emit 16 or 23 channels depending on fallback).
- **Dual pooling (Mean + Max)** — mean captures global activity level; max captures peaky ictal bursts. Concatenation doubles the feature dimension fed to the classifier.
- **Dropout 0.3 between LSTM layers** — primary regulariser. Weight decay is the secondary one.

VanillaLSTM (`vanilla_lstm.py:47–122`)

- 2-layer unidirectional LSTM, hidden 128, dropout 0.3.
- Acts as a reference floor: no bidirectionality, no self-attention, no CNN frontend.

BiLSTM (`bilstm.py:44–113`)

- 2-layer bidirectional LSTM; forward + backward hidden states concatenated → 256-dim per timestep.
- Within a 1-s window, the "future" is visible at training and inference — a reasonable assumption for offline seizure detection and a small assumption for clinical systems that buffer a few seconds of signal.

AttentionBiLSTM (`attention_bilstm.py:19–130`)

- BiLSTM → **learnable positional embedding (trunc-normal init)** → 4-head self-attention with residual → pool.
- The positional embedding is important: once we add self-attention on top of an RNN, the attention block itself is permutation-invariant, so position must be re-injected.

FeatureBiLSTM (`feature_bilstm.py:19–139`)

- Unlike the other LSTMs, this consumes **226 hand-crafted features per window over a sequence of 10 windows** (i.e. 10 s context). A residual projection 226 → 256 → 128 stabilises training under very small effective batch sizes, and a 4-head temporal MHA learns which of the 10 windows is most discriminative.
- Intended as a "tabular-features" baseline alongside the raw-waveform models.

CNNLSTM (`cnn_lstm.py:74–186`)

- Three parallel conv branches with kernel sizes **3 / 15 / 31** mimic the three clinical EEG time-scales: fast spikes, spike-wave complexes, and slow ictal rhythms.
- Branch A (32 ch, k = 3, stride = 2) + Branch B (64 ch, k = 15, stride = 2) + Branch C (64 ch, k = 31, stride = 1, then downsample) → concat to 160 ch → BiLSTM → MHA → dual pool.
- This is the architecture that produces the strongest baseline numbers (Section 8).

3.2 Benchmark suite `M1` – `M6`

`lstm_benchmark_models/architectures/m[1–6]_*.py` reproduces the five LSTM architectures above in a standardised shape, plus one novel graph variant:

ID	Key addition over baseline
M1	Baseline VanillaLSTM with `ChannelAttention(reduction=4)`
M2	BiLSTM with combined channel + spatial attention
M3	`CrissCrossBiLSTM` — attention applied horizontally (time) and vertically (channels) — cheaper than full 2D self-attention
M4	CNN-LSTM with SE blocks at each conv
M5	Feature-BiLSTM (226-feature input)
M6	**Graph BiLSTM** — channels as graph nodes, GAT-style attention over channel connectivity

M1–M6 share `_build_input_proj`, `_build_lstm`, and `_build_classifier` helpers so ablations can be run cleanly.

3.3 HierarchicalLSTM (`improved_lstm_models/.../hierarchical_lstm.py:60–136`)

Two-level model operating on **60-window (60-second) sequences**: 1. Window encoder (per-window CNN) squashes (16, 256) → 128. 2. BiLSTM over the sequence of 60 128-dim window embeddings. 3. Attention pooling over the 60 windows (learnable softmax weights) → classifier.

Rationale: seizures have pre-ictal build-up that a 1-s window cannot capture. HierarchicalLSTM gives the model a minute of context while avoiding the $O(T^2)$ cost of self-attention over 15 360 raw samples.

3.4 M7 VQ-Transformer (`ensemble_transformers/.../m7_vq_transformer.py:13–104`)

- Patch embedding via `unfold(32)` → 8 patches × 512-dim → linear 512→128.
- Learned positional encoding (1, 9, 128) with CLS token.
- 4-layer 4-head Transformer encoder (GELU, FFN 128→512→128).
- **Vector quantiser with 512 codes** injects a discrete bottleneck, producing a VQ loss (commitment + codebook). This is the only model in the suite that learns a discrete "EEG vocabulary", giving a handle on interpretability (each patch is mapped to a code index).

3.5 EEGMamba & EEGMambaMoE (`hugging_face_mamba_moe/.../eeg_mamba.py:21–80`)

Bidirectional Mamba blocks (forward + backward selective state-space) with `d_model = 128`, `d_conv = 4`, a `d_state = 16` SSM, and residual connections. Global mean pooling → LayerNorm → classifier. The MoE variant adds a router over expert FFNs.

Engineering caveats (see §7.2 and §11): early versions suffered NaN losses due to (a) a `.expand()` in the SSM output that 256× amplified gradients (fixed in commit `9fbfe5f`), (b) an unstable log-space scan (replaced by a `torch.jit.script` compiled sequential scan in `7c60642`), and then (c) a 10–50× speed-up from a vectorised parallel scan in `94a5b2b`.

3.6 Pretrained foundation models

- **BENDR** (`pretrained/bendr.py:11–60`) — Braindecode's `braindecode/braindecode-bendr` backbone consumed at 16 ch × 256 samples × 256 Hz. Per-channel z-score normalisation is **mandatory at input** (fix in commit `6516657`) or the pretrained distribution shift causes rapid divergence. `freeze_backbone=True` is supported for linear-probe evaluation.
- **EEGPT** (`pretrained/eegpt.py:11–57`) — `braindecode/eegpt-pretrained` backbone. Pretrained on 64 channels; we override `chs_info=None` (same commit `6516657`) so the model does not enforce its 64-channel channel schema, allowing 16-channel fine-tuning.
- `approach2/modules/pretrained_encoders.py` additionally exposes **CBraMod (ICLR 2025)**, **BIOT**, **LaBraM (ICLR 2024)** — wired as optional backbones for the M1–M7 variants in `approach2/architectures/`.

3.7 TinySeizureNet (`approach3/architectures/tiny_seizure_net.py`)

Depthwise-separable 1-D CNN (`DepthwiseSeparableConv1d`) with < 1 M parameters, trained via `MultiTeacherDistillation` from an ensemble of M1–M6 teachers. Intended for on-device deployment; a `QuantizedTinyNet` variant is provided for post-training quantisation.

3.8 Loss & attention utilities

- `src/models/utils/losses.py` : `FocalLoss` (§5) — single source of truth after the Apr-14 consolidation.
- `src/models/utils/metrics.py` : shared `compute_f1_score`, `compute_auc_roc`, `compute_sensitivity`, `compute_specificity`, plus the Youden-J threshold finder at lines 115–142.
- Attention modules: `ChannelAttention`, `ChannelAttentionWithContext`, `SpatialChannelAttention` (`lstm_benchmark_models/modules/channel_attention.py`), `CrissCrossAttention` (`.../criss_cross_attention.py`), `GraphAttention` (`.../graph_attention.py`).

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4. Why LSTM? Design Rationale & Alternatives

LSTM as default inductive bias. EEG is an ordered, locally-stationary time series sampled at 256 Hz. For a 1-second window (256 timesteps) a 2-layer LSTM has an effective receptive field well beyond the window length and is cheap enough to run on commodity GPUs. Recurrence is a strong inductive bias for signals whose instantaneous meaning depends on short-term history — precisely the case for ictal rhythm onset.

Why bidirectional. Offline seizure detection lets us look at the full window before deciding; a BiLSTM therefore sees both onset and offset of a transient spike-wave complex. The per-timestep embeddings are concatenated (256-dim), doubling classifier capacity without doubling recurrence length.

Why dual (mean + max) pooling. Seizures often manifest as localized bursts in time. Mean pooling alone smooths these away; max pooling alone is noise-sensitive. The concatenation is a cheap hedge.

Why add a CNN frontend (CNN-LSTM). Multi-scale kernels (3 / 15 / 31 samples ≈ 12 / 59 / 121 ms at 256 Hz) directly correspond to clinical EEG band interpretation (high-frequency spikes, spike-wave complexes, slow rhythms). CNN-LSTM is the baseline's top performer (Section 8), validating that temporal inductive bias + local translation invariance complement each other.

Why also try Transformers. Pure self-attention is permutation-invariant and data-hungry, but once we have the VQ bottleneck (M7) it doubles as an interpretable representation learner. The VQ codebook gives each 32-sample patch a discrete identity, which is useful for downstream retrieval / anomaly flagging.

Why also try Mamba / SSM. Mamba scales linearly in sequence length, which matters for the hierarchical 60-s context setting (3840 1-s window embeddings or 15 360 raw samples). The selective state-space mechanism is well-suited to signals with long quiet stretches punctuated by bursts.

Why pretrained (BENDR / EEGPT / CBraMod / LaBraM). CHB-MIT is small (23 subjects). Pretrained foundation models bring self-supervised priors learned from orders of magnitude more EEG, providing a strong feature extractor with only a linear probe fine-tuned. This is the standard "few-shot medical ML" recipe.

Compared to classical ML (SVM / RF on spectral features). FeatureBiLSTM is our nod to this family: it accepts 226 engineered features per window. We do not train a separate SVM/RF; the feature-based BiLSTM provides an implicit upper bound on what a classical pipeline could reach because it has

access to the same features plus a temporal model on top.

5. Loss Function & Class Imbalance

The training loss is a numerically-stable Focal Loss with post-hoc `pos_weight`:

$$ \mathcal{L}_{\text{focal}}(p,y) \;=\; -\,\alpha_t\,(1 - p_t)^{\gamma}\,\log p_t \;\cdot\; w^{\,y} $$

with $p_t = p$ if $y = 1$ else $1 - p$, $\alpha_t = \alpha$ if $y = 1$ else $1-\alpha$, $w_{\text{pos}} = n_-/\max(n_+,1)$, and y smoothed by $\epsilon \in [0.02, 0.1]$ before passing to the loss.

Defaults: $\alpha = 0.25$, $\gamma \in \{1.0, 2.0\}$, `pos_weight` computed from the training window index at training start.

Critical implementation fixes (visible in git history):

Commit	Fix
`f6460ac`	Inverted `pos_weight` (was `n_+ / n_-`, should be `n_- / n_+`) corrected; legacy results pre-dating this should be regarded with caution.
`0026241`	`pos_weight` must be applied as a flat multiplier **after** focal modulation, not inside the BCE call. Inside-BCE semantics combine with the focal factor in a way that double-weights hard positives.
`66d6624`	Cast `FocalLoss` inputs to `float32` to prevent FP16 underflow under `torch.amp` autocast. Without this, `(1-p_t)^\gamma` flushes to zero when `p_t \to 1`, yielding zero gradient on the hard negatives the loss is meant to emphasise.
`d9bedbc`	Target threshold check `targets > 0.5` rather than `targets == 1` — needed once label smoothing was added.

Ref: `src/models/utils/losses.py` (consolidated FocalLoss). `approach2/train.py:489` uses $\gamma = 1.0$, $\epsilon = 0.02$; `train_hf.py:39–67` uses $\gamma = 2.0$, $\alpha = 0.25$ as CLI defaults.

6. Hyperparameters & Optimisation

6.1 Training-script inventory

Script	Models	Epochs	Batch	LR	WD	Patience	Notes
`lstm_benchmark_models/train_baseline.py`	M1–M6	30	64	1e-3	1e-4	7	Focal γ = 2
`improved_lstm_models/train.py`	HierarchicalLSTM	100	32	5e-5	1e-4	20	After bbe24e0 / 9c8b7c7
`ensemble_transformers/train_ensemble.py`	M1–M7 ensemble	50	64	5e-4	1e-4	15	Then ensemble voting
`hugging_face_mamba_moe/train_hf.py`	Mamba / MoE / BENDR / EEGPT / etc. (12 models)	20	32	1e-3	1e-4	7	`--model all`, `--grad_clip_norm 1.0`, `--decision_threshold 0.35`
`approach2/train.py`	M1–M7 + pretrained backbones	50	64	5e-4	1e-4	15	WarmupCosineScheduler
`approach3/train_mamba.py`	EEGMamba + MoE	20	32	1e-3	1e-4	7	—

All scripts share the same YAML master config at `src/models/config.yaml` (augmentation probabilities, window sec, stride sec, seizure ratio, etc.).

6.2 Optimiser and schedule

Every script uses **AdamW** (`torch.optim.AdamW`) with weight decay `1e-4`. E.g. `approach2/train.py:377`, `train_hf.py:79`.

The default schedule is `WarmupCosineScheduler` (custom, `approach2/train.py:90–110` and `improved_lstm_models/train.py:34–75`):

$$ \eta_t \;=\; \begin{cases} \eta_{\max}\cdot \dfrac{t+1}{T_{\text{warm}}}, & t < T_{\text{warm}} \\ \eta_{\min} + \tfrac{1}{2}(\eta_{\max}-\eta_{\min})\Bigl(1 + \cos\bigl(\pi\cdot \tfrac{t - T_{\text{warm}}}{T - T_{\text{warm}}}\bigr)\Bigr), & t \ge T_{\text{warm}} \end{cases} $$

with $T_{\text{warm}} \in \{5,10\}$, $\eta_{\min} = 10^{-6}$. An alternate scheduler appears in `compare.py:165` (`CosineAnnealingWarmRestarts`, $T_0 = T - T_{\text{warm}}$) and in the legacy `improved/train_regularized.py:69` (`ReduceLROnPlateau`, factor 0.5, patience 3).

6.3 Regularisation and stability

- **Dropout** 0.3 (LSTM / BiLSTM); 0.1 (Transformer encoder).
- **Label smoothing** $\epsilon \in \{0.02, 0.1\}$ — applied before the loss.
- **Gradient clipping** by L2 norm = 1.0 (`train_hf.py:182`).
- **Mixed precision** via `torch.amp.autocast` with float32 loss casts (see fix `66d6624`).

6.4 Tuning strategy

Tuning in this codebase is **manual + iterative**, not grid/random search. Each significant hyperparameter change is traceable to a commit message identifying the failure mode it addresses (e.g., `9c8b7c7 — lower improved_lstm lr to 0.0001 + more warmup/patience to fix unstable val loss`; `71277d3 — lower focal_gamma=1.0, higher lr=3e-3 to prevent loss collapse`). The git log therefore *is* the tuning log. Section 11 condenses it.

6.5 Early stopping and threshold tuning

- **Early stop on val F1** (not val loss) — switched in commits `bd32418` / `bbe24e0`. Compare uses `val_f1 > best + 1e-4`; patience 7 (HF), 15 (ensemble), 20 (improved LSTM). `improved_lstm_models/train.py:116-139`.
- **Threshold tuning**
- Youden's J: `src/models/utils/metrics.py:115-142`, scan [0.01, 0.99] step 0.01, maximise $J = \text{sens} + \text{spec} - 1$.
- F1-max: `train_hf.py:199-216` and `ensemble_hf.py`, scan [0.05, 0.95] step 0.05. Default decision threshold (pre-tuning) = 0.35 (`train_hf.py:60`).

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7. Training Strategy & Iteration

7.1 Cross-cutting choices

- **Subject-independent splits** from `core/stratify.py` prevent the easiest kind of leakage. Data augmentation runs per-sample inside the dataset, so augmented copies of a subject cannot appear in both train and val.
- **Balanced training, unbalanced evaluation.** Oversampling is enabled only on train (`labels.py:184-210`); val / test keep natural prevalence for realistic sensitivity/specificity numbers.
- **Mixed precision** everywhere (AMP autocast). Loss is cast to float32 before the focal modulation (commit `66d6624`).
- **Gradient clipping** norm = 1.0 as a blanket safeguard against exploding gradients in RNN / SSM blocks.

7.2 Stability engineering (SSM / Mamba)

Two Mamba-specific fixes are worth surfacing because they are not obvious from the API: - Commit `9fbfe5f` removes a `.expand()` in the SSM output whose broadcast shape caused a 256× gradient amplification during back-prop, producing NaN losses within a handful of iterations. - Commit `7c60642` replaces a hand-written log-space parallel scan with a `torch.jit.script`-compiled sequential scan, which sacrifices some throughput for numerical stability. A follow-up, `94a5b2b`, re-introduces a vectorised parallel scan that is 10–50× faster than the scripted loop while preserving stability.

7.3 Ensemble evaluation

`src/models/hugging_face_mamba_moe/ensemble_hf.py` (introduced in `bd32418`) aggregates probabilities from multiple trained HF models by **mean voting** or **weighted voting**, tunes the ensemble's operating threshold via the same F1-max routine, and computes confusion-matrix metrics. Temporal smoothing across adjacent windows (moving average or consecutive-run minimum) is supported.

7.4 Distillation (Approach 3)

`MultiTeacherDistillation` (`approach3/architectures/tiny_seizure_net.py`) trains `TinySeizureNet` against the soft outputs of M1–M6, combining a KL-divergence term on logits with a CE term on labels. This is the fastest path to a < 1 M-parameter deployable model without sacrificing much sensitivity.

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8. Results & Evaluation

8.1 Metric set (from `src/models/utils/metrics.py`)

- F1 (sklearn).
- AUROC (sklearn `roc_auc_score`).
- Sensitivity = TP / (TP + FN).
- Specificity = TN / (TN + FP).
- Precision (inline in train scripts).
- Confusion matrix (TP, TN, FP, FN) as absolute counts.
- Threshold value at which those metrics are reported (for reproducibility).

8.2 Persisted baseline results

File: `src/models/legacy_baseline/results/baseline_results.json` (verbatim, all numbers rounded to 4 dp):

Model	Accuracy	Sensitivity	Specificity	F1	AUC	Train time (s)
Vanilla LSTM	0.6447	0.3139	0.7865	0.3464	0.5629	3 401
BiLSTM	0.6825	0.2598	0.8637	0.3293	0.6114	6 488
Attention-BiLSTM	0.6926	0.2731	0.8724	0.3476	0.6406	7 501
CNN-LSTM	**0.6819**	**0.5692**	**0.7302**	**0.5177**	**0.7118**	**2 143**

Reading the table. - Pure LSTM / BiLSTM achieve higher **specificity** than CNN-LSTM but fail to catch most seizures — sensitivity is 0.26–0.31, which is clinically unacceptable. - CNN-LSTM roughly doubles sensitivity (0.57) while retaining 0.73 specificity. F1 jumps from ~0.34 to ~0.52 and AUC from ~0.56–0.64 to 0.71. - Training time for CNN-LSTM (2 143 s) is the **fastest** of the four — the convolutional downsampling (stride 2) shrinks the LSTM's input length by 2×, and the conv stack is GPU-efficient compared to pure recurrence. - This single comparison is the strongest narrative hook for the paper: *multi-scale temporal convolution is the single most valuable frontend for seizure detection in this regime*, and motivates the subsequent adoption of richer frontends (M4 SE-CNN, VQ-Transformer patch embed, pretrained encoders).

8.3 Other models (checkpoints vs persisted metrics)

Current repository state: `outputs/models/*_best.pt` checkpoints are produced by every training script, and per-epoch history is logged to `logs/history.csv` and `logs/test_metrics.json` (`train_hf.py:141, 157`). However, the only numeric test-set results committed to disk are the four rows above. **Results for M1–M7, HierarchicalLSTM, EEGMamba, BENDR, EEGPT, TinySeizureNet, and the ensembles are not persisted as JSON in this branch** and need to be re-extracted from the checkpoints before inclusion in a camera-ready paper.

Action for authors: run `train_hf.py --model all --data_path ... --resume` (to load best checkpoints) and collect the `test_metrics.json` per model into one table before submission.

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9. EEG Classification & Detection Context

- **Temporal dependencies.** Seizures evolve over milliseconds (spike, spike-wave) to seconds (ictal rhythm build-up). The 1-s window + 2-layer LSTM spans ~256 samples of context, enough to represent a full spike-wave complex (~1–3 Hz). HierarchicalLSTM lifts this to 60 s so the model can see pre-ictal precursors. Mamba's linear-time scaling makes longer windows tractable.
- **Channel semantics.** Focal seizures involve a small subset of electrodes; generalised seizures involve all. ChannelAttention learns per-example channel gating, mirroring a clinician's habit of focusing on the leads where the signal is abnormal. Graph attention in M6 goes further: adjacent electrodes have correlated activity, and a learned graph can encode that.
- **Multi-scale kernels.** The 3 / 15 / 31-sample kernels in CNN-LSTM correspond roughly to the γ / β / α–θ band range at 256 Hz — a standard band decomposition built into the inductive bias of the convolutions instead of a preprocessing FFT.
- **Subject independence.** Stratified subject splits are the right protocol for clinical deployment: the model must generalise to unseen patients. Oversampling is kept to the train split so val/test specificity reflects real-world false-alarm behaviour.
- **Robustness.** Augmentations (time warp, noise, magnitude scale, time shift) emulate the kinds of variation a real recording session introduces (electrode impedance drift, amplifier gain drift, sampling-clock jitter).

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10. Comparative Insights & Future Work

Empirical takeaways from the current repo state. 1. Adding a multi-scale CNN frontend produces the single biggest measured quality jump. 2. Pure self-attention (without VQ or pretraining) is not attempted at scale; the Transformer-flavoured experiments wear either a VQ bottleneck (M7) or a pretrained backbone (BENDR / EEGPT). 3. State-space models (Mamba) required substantial numerical-stability work but are now trainable; their value proposition — long context with linear cost — is only realised by the hierarchical / 60-s paths. 4. Loss-function semantics matter: the sequence `f6460ac → 0026241 → 66d6624 → d9bedbc` systematically corrected Focal Loss to behave as intended.

Future directions. - **Longer windows**: scale the hierarchical path from 60 s to 5 min; compare against a plain Mamba over raw 15 000-sample sequences. - **Seizure-type multi-class**: the label machinery (`core/labels.py`) already supports keyword-based multi-class tagging (focal, generalised, absence). Enabling this is a one-line config change plus a modified classifier head. - **Patient-specific fine-tuning**: train a generic model on 20 patients, fine-tune the last layer on the first 30 minutes of the held-out patient; compare to fully subject-independent evaluation. - **Online / streaming deployment**: `TinySeizureNet` + `QuantizedTinyNet` gives a < 1 MB model; pair with a causal inference loop and a moving-average smoother for false-alarm control. - **Calibration**: all current models are evaluated at a tuned operating threshold. Adding temperature scaling and Brier-score reporting would make the probability outputs clinically trustworthy. - **Unified results harness**: persist a `results/{model_name}.json` for every model so the next version of this document can replace §8.3 with a full comparison table.

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11. Timeline of Modeling Code Evolution

Reconstructed from `git log --oneline` . Oldest work is at the bottom; most recent at the top. We group into four phases.

Phase 4 — Ensembling, stability polish, pretrained fixes (current, mid-April 2026)

Commit	Change
`6516657`	**HF train**: robust best-F1 tracking with NaN guard; **BENDR** z-score normalise at input; **EEGPT** `chs_info=None` to unlock 16-channel fine-tuning.
`bbe24e0`	**Improved-LSTM**: early stop on val F1 (not loss); reduce model size; lower LR to 5e-5.
`bd32418`	Introduce `ensemble_hf.py` : mean / weighted voting + threshold tuning.
`7c60642`	Replace unstable log-space scan with `torch.jit.script` -compiled sequential scan in Mamba (removes NaN).
`9fbfe5f`	Remove `.expand()` in SSM output — eliminates 256× gradient amplification → removes NaN loss.
`9c8b7c7`	Lower `improved_lstm` LR to 1e-4, more warmup/patience.
`94a5b2b`	Replace sequential Python SSM loop with vectorised parallel scan (10–50× speed-up).
`3d5a27d`	`train_hf` default tuning (focal loss, threshold tuning, grad-clip); add `--model all` .
`21de52a`	`--data_path` flag on `train_hf.py` to load `.pt` tensors directly.
`a33506e`	Project-root-relative paths in `data_loader` config.
`30055d1`	Drop unsupported `config_path` kwarg from `get_dataloaders()` .
`fd2ca90`	`run_all_models.py` — one-command training across all dirs.
`ad7183a`	Filter kwargs per model to avoid unexpected-keyword errors.
`b13c5dd`	`--model all` across benchmark.
`8342be0`	Cross-cutting bugfix / perf / robustness sweep over all 4 model dirs.
`d761539`	Split hygiene: val/ for early stopping, test/ for final eval.
`db3a9bc`	Fix ensemble-weights device mismatch; `approach2` warmup scheduler.

Phase 3 — Architectural diversification (Approach 2 / Approach 3, early April 2026)

Commit	Change
`605ac69`	**Fix approach2 training collapse** (4 bugs).
`d9bedbc`	FocalLoss targets > 0.5 check; graph-attention reshape.
`0026241`	Apply `pos_weight` as flat multiplier after focal modulation (not inside BCE).
`66d6624`	Cast FocalLoss inputs to float32 to prevent FP16 underflow in AMP.
`fa1961e`	Plan doc: approach2 training-collapse fix plan.
`efe1d9f`	Fix 10 bugs in `improved_lstm` : crashes, logic errors, dead code.
`844dbfc`	Fix relative imports in `improved_lstm/train.py` .
`44c4903`	**Add `improved_lstm`** with augmentation, ensemble, enhanced training.
`de26605`	Fix `attention_bilstm` (remove downsampling); `feature_bilstm` (3D shape extraction).
`f6460ac`	Fix inverted `pos_weight` ; add tensors format support; deprecated `torch.amp` API.
`71277d3`	Lower focal_gamma to 1.0, raise LR to 3e-3 to prevent loss collapse.
`183ddbb`	Fix UnboundLocalError: use `X_train/X_test` instead of `X_raw` for attention_bilstm.
`cb55216`	Fix Mamba block tensor shape: `torch.stack` vs `torch.cat` .
`093780a`	Support loading real EEG from tensors (`data.pt` , `labels.pt`).
`7870440`	Fix format-string TypeError on None values.
`4370e93`	Remove broken `fn_multiplier` ; fix import paths in approach2/3.
`785a2e5`	Fix `run_benchmark.py` format bug; add training utils.
`13f0986`	**Merge**: Add Approach 2 and Approach 3 models.
`f794e93`	**Add Approach 2** (7 models + pretrained encoders) and **Approach 3** (Mamba, Diffusion, Pre-ictal).

Phase 2 — Baseline improvements (LSTM-first era)

Commit	Change
`d62d977`	Move old baseline scripts to `legacy_baseline/`.
`4ef865c`	Add comprehensive summary of LSTM improvements.
`ad959af`	**Complete LSTM improvements**: parallel multi-scale CNN, temporal attention, ensemble inference, training optimisations.
`0c69cad`	Implement Focal Loss, FN weighting, label smoothing, LR warmup, ChannelAttention, positional encoding.

Phase 1 — Baseline (pre-mid-April)

Baseline M1–M6 LSTM benchmark suite, initial FocalLoss, legacy `dataloaders/`, and the `baseline_results.json` numbers reported in §8.2.

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12. References & Datasets

- **CHB-MIT Scalp EEG Database**, PhysioNet. (Shoeb & Guttag.)
- **Focal Loss**: Lin et al., *Focal Loss for Dense Object Detection*, ICCV 2017.
- **BENDR**: Kostas et al., *BENDR: Using Transformers and a Contrastive Self-Supervised Learning Task to Learn from Massive Amounts of EEG Data*.
- **EEGPT**: *EEGPT: Pretrained Transformer for Universal and Reliable Representation of EEG Signals*, NeurIPS 2024.
- **LaBraM**: *Large Brain Model for Learning Generic Representations with Tremendous EEG Data in BCI*, ICLR 2024.
- **CBraMod**: ICLR 2025.
- **BIOT**: *BIOT: Cross-data Biosignal Learning in the Wild*.
- **Mamba**: Gu & Dao, *Mamba: Linear-Time Sequence Modeling with Selective State Spaces*, 2023.

Appendix A — File-Level Citation Index

For any claim in this document, the cited file:line can be opened directly. Primary references:

- Data pipeline: `src/data_loader/core/{signal.py, channels.py, labels.py, stratify.py, augment.py, bids.py, io.py, download.py}`, `src/data_loader/dataset/{base.py, loaders.py, factory.py}`, `src/data_loader/config.yaml`.
- Architectures: `src/models/architectures/{vanilla_lstm.py, bilstm.py, attention_bilstm.py, feature_bilstm.py, cnn_lstm.py}`, `src/models/lstm_benchmark_models/architectures/m[1-6]_*.py`, `src/models/improved_lstm_models/architectures/hierarchical_lstm.py`, `src/models/ensemble_transformers/architectures/m7_vq_transformer.py`, `src/models/hugging_face_mamba_moe/architectures/eeg_mamba.py`, `src/models/hugging_face_mamba_moe/architectures/pretrained/{bendr.py, eegpt.py}`, `src/models/approach3/architectures/tiny_seizure_net.py`.
- Training / loss / metrics: `src/models/utils/{losses.py, metrics.py}`, `src/models/approach2/train.py`, `src/models/hugging_face_mamba_moe/{train_hf.py, ensemble_hf.py}`, `src/models/improved_lstm_models/train.py`, `src/models/lstm_benchmark_models/train_baseline.py`, `src/models/ensemble_transformers/train_ensemble.py`.
- Persisted results: `src/models/legacy_baseline/results/baseline_results.json`.
- Master config: `src/models/config.yaml`.

End of draft. Next revision should populate §8.3 with per-model test metrics extracted from the `outputs/models/*_best.pt` checkpoints.